

OCR Further Mathematics
Pure Core - Complex Numbers
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1 Use de Moivre's theorem to find the constants A, B and C in the identity $\sin^5 \theta \equiv A \sin \theta + B \sin 3\theta + C \sin 5\theta$. [4]

2 A function $f(z)$ is defined on all complex numbers z by $f(z) = z^3 - 3z^2 + kz - 5$ where k is a real constant. The roots of the equation $f(z) = 0$ are α, β and γ . You are given that $\alpha^2 + \beta^2 + \gamma^2 = -5$.

Explain why $f(z) = 0$ has only one real root. [3]

3(a) Sketch on a single Argand diagram the loci given by

(i) $|z - 1 + 2i| = 3$, [2]

(ii) $|z + 1| = |z - 2|$. [2]

(b) Indicate, by shading, the region of the Argand diagram for which $|z - 1 + 2i| \leq 3$ and $|z + 1| \leq |z - 2|$. [2]

4(a) In this question you must show detailed reasoning.

The complex numbers z_1 and z_2 are given by $z_1 = 3 - 7i$ and $z_2 = 2 + 4i$.

Express each of the following as exact numbers in the form $a + bi$.

(i) $3z_1 + 4z_2$ [1]
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(ii) $z_1 z_2$ [2]

(iii) $\frac{z_1}{z_2}$ [2]

(b) Write z_1 in modulus-argument form giving the modulus in exact form and the argument correct to 3 significant figures. [3]

5(a) By writing $\sin\theta$ in terms of $e^{i\theta}$ and $e^{-i\theta}$ show that

$$\sin^6\theta = \frac{1}{32}(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta).$$

[5]

(b) Hence show that $\sin\frac{1}{8}\pi = \frac{1}{2}\sqrt[6]{20 - 14\sqrt{2}}$.

[3]

6(a) In an Argand diagram the points representing the numbers $2 + 3i$ and $1 - i$ are two adjacent vertices of a square, S.

Find the area of S.

[3]

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(b) Find all the possible pairs of numbers represented by the other two vertices of S.

[4]

7(a) In this question you must show detailed reasoning.

The complex number $-4 + i\sqrt{48}$ is denoted by z .

Determine the cube roots of z , giving the roots in exponential form.

[6]

(b) The points which represent the cube roots of z are denoted by A , B and C and these form a triangle in an Argand diagram.

Write down the angles that any lines of symmetry of triangle ABC make with the positive real axis, justifying your answer.

[3]

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8 In this question you must show detailed reasoning.

Solve the equation $4z^2 - 20z + 169 = 0$. Give your answers in modulus-argument form.

[5]



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9(a) You are given that the cubic equation $2x^3 + px^2 + qx - 3 = 0$, where p and q are real numbers, has a complex root $\alpha = 1 + i\sqrt{2}$.

Write down a second complex root, β .

[1]



(b) Determine the third root, γ .

[2]

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(c) Find the value of p and the value of q .

[2]

(d)

Show that if n is an integer then $\alpha^n + \beta^n + \gamma^n = 2 \times 3^{\frac{1}{2}n} \times \cos n\theta + \frac{1}{2^n}$ where $\tan \theta = \sqrt{2}$.

[4]

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10

By expanding $\left(z^2 + \frac{1}{z^2}\right)^3$, where $z = e^{i\theta}$, show that $4 \cos^3 2\theta = \cos 6\theta + 3 \cos 2\theta$.

[5]



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11(a) In this question you must show detailed reasoning.

Determine the square roots of $25i$ in the form $r e^{i\theta}$, where $0 \leq \theta < 2\pi$.

[3]



(b) Illustrate the number $25i$ and its square roots on an Argand diagram. [1]

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12(a) You are given the complex number $\omega = \cos \frac{2}{5}\pi + i \sin \frac{2}{5}\pi$ and the equation $z^5 = 1$.

Show that ω is a root of the equation.

[2]

(b) Write down the other four roots of the equation.

[1]

(c) Show that $\omega + \omega^2 + \omega^3 + \omega^4 = -1$.

[2]

(d)

Hence show that $\left(\omega + \frac{1}{\omega}\right)^2 + \left(\omega + \frac{1}{\omega}\right) - 1 = 0$.

[3]

(e) Hence determine the value of $\cos \frac{2}{3}\pi$ in the form $a + b\sqrt{c}$ where a, b and c are rational numbers to be found. [4]



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- 13 You are given that $x = 2 + 5i$ is a root of the equation $x^3 - 2x^2 + 21x + 58 = 0$.

Solve the equation.

[4]

- 14 Indicate by shading on an Argand diagram the region

$$\{z : |z| \leq |z - 4|\} \cap \{z : |z - 3 - 2i| \leq 2\}.$$

[3]

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15 In this question you must show detailed reasoning.

You are given that $z = \sqrt{3} + i$.

n is the smallest positive whole number such that z^n is a positive whole number.

(a) Determine the value of n .

[3]



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(b) Find the value of z^n .

[1]

You are given that the matrix $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{2a-a^2}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix}$, where a is a positive constant, represents the

transformation R which is a reflection in 3-D.

(a) State the plane of reflection of R . [1]

(b) Determine the value of a . [3]

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(c) With reference to R explain why $A^2 = I$, the 3×3 identity matrix.

[2]



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The loci C_1 and C_2 are given by $|z - 1| = 5$ and $\arg(z + 4 + 4i) = \frac{1}{4}\pi$ respectively.

(a) Sketch on a single Argand diagram the loci C_1 and C_2 .

[4]



(b) Indicate by shading on your Argand diagram the following set of points.

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$$\{z: |z - 1| \leq 5\} \cap \{z: 0 \leq \arg(z + 4 + 4i) \leq \frac{1}{4}\pi\}$$

[2]

18 In this question you must show detailed reasoning.

For the complex number z it is given that $|z| = 2$ and $\arg z = \frac{1}{6}\pi$.

Find the following in the form $a + ib$, where a and b are exact numbers.

(a) z [2]

(b) z^2 [2]

(c) $\frac{z}{z^*}$ [3]

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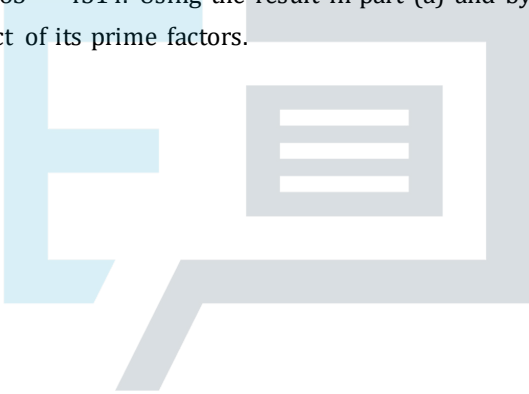
19 In this question you must show detailed reasoning.

(a) Express $(6 + 5i)(7 + 5i)$ in the form $a + bi$.

[2]

(b) You are given that $17^2 + 65^2 = 4514$. Using the result in part (a) and by considering $(6 - 5i)(7 - 5i)$ express 4514 as a product of its prime factors.

[4]



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20 In this question you must show detailed reasoning.

Find the square roots of $24 + 10i$, giving your answers in the form $a + bi$.

[5]



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21 In this question you must show detailed reasoning.

(a) Show that $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$.

[1]

(b) Hence, show that $\frac{2}{e^{2i\theta} - 1} = -(1 + i \cot \theta)$.

[3]

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(c) Two series, C and S , are defined as follows.

$$C = 2 + 2 \cos \frac{\pi}{10} + 2 \cos \frac{\pi}{5} + 2 \cos \frac{3\pi}{10} + 2 \cos \frac{2\pi}{5}$$

$$S = 2 \sin \frac{\pi}{10} + 2 \sin \frac{\pi}{5} + 2 \sin \frac{3\pi}{10} + 2 \sin \frac{2\pi}{5}$$

By considering $C + iS$, find a simplified expression for C in terms of only integers and $\cot \frac{\pi}{20}$.

[8]



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(d) Verify that $S = C - 2$ and, by considering the series in their original form, explain why this is so.

[2]

22 The complex number $2 + i$ is denoted by z .

(a) Show that $z^2 = 3 + 4i$.

[2]

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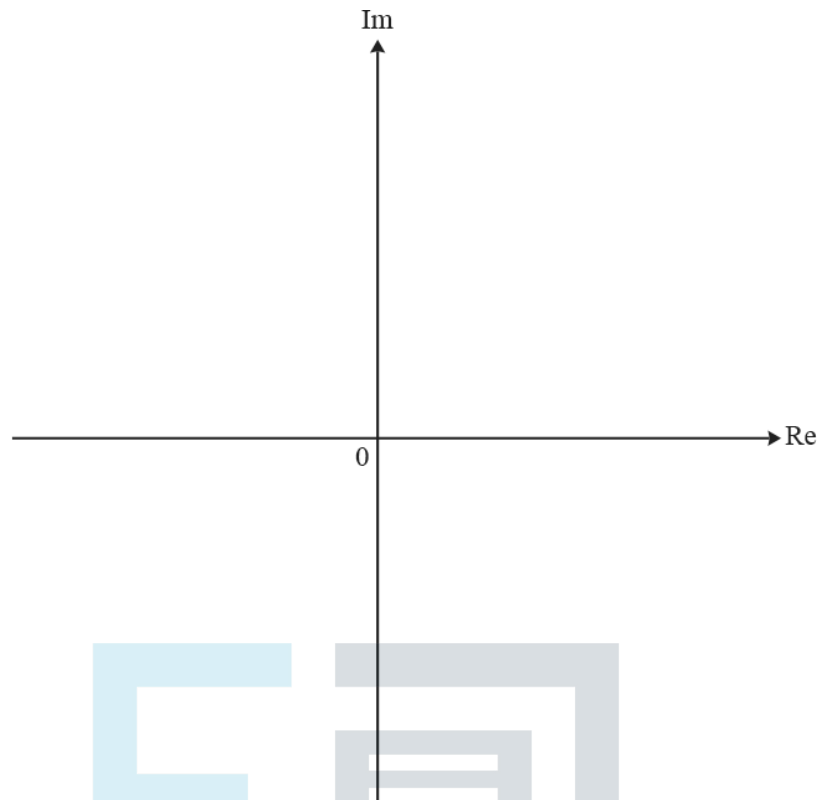
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(b) Plot the following on the Argand diagram below.

- z
- z^2

[1]



(c) State the relationship between $|z^2|$ and $|z|$.

[1]

(d) State the relationship between $\arg(z^2)$ and $\arg(z)$.

[1]

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- 23 (i) By expressing $\cos \theta$ in terms of $e^{i\theta}$ show that

[4]

$$\cos^6 \theta = \frac{1}{32}(\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10).$$

- (ii) Hence solve, for $0 \leq \theta \leq \pi$,



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$$\cos 6\theta + 6 \cos 4\theta + 2 \cos 2\theta = 3.$$

[5]



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- 24 In an Argand diagram the points O , A and B are represented by the complex numbers 0 , z and $2e^{\frac{1}{3}\pi i}z$ respectively, where z is a complex number with modulus 5.
- (i) Calculate the exact area of the triangle OAB . [3]

The numbers $-1 + i$ and $3 + 3i$ are represented by the points P and Q respectively. The complex number w is represented by the point R , such that $PQ = PR$ and angle $QPR = \frac{1}{4}\pi$.

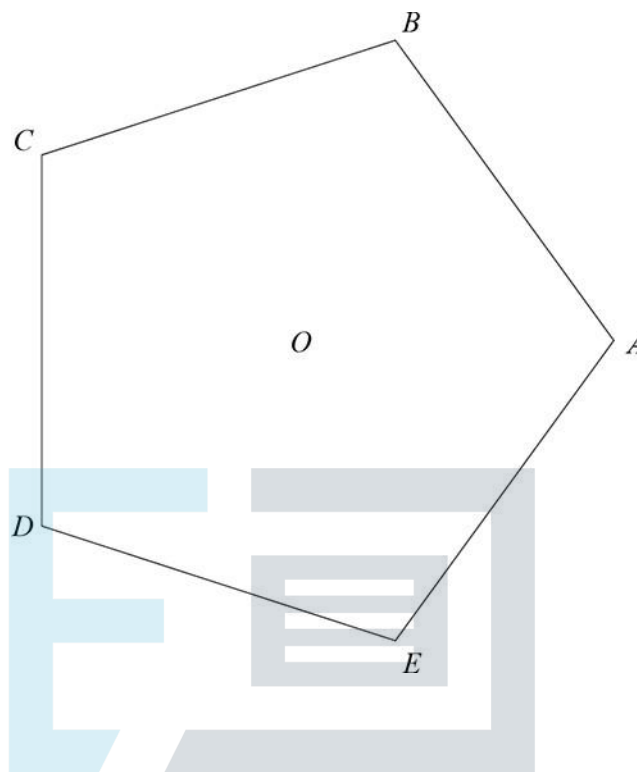
- (ii) Sketch an Argand diagram showing P , Q and the two possible positions of R . Calculate the possible values of w , giving your answers in the form $a + bi$. [5]

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- 25 The Argand diagram below shows the origin O and pentagon $ABCDE$, where A, B, C, D and E are the points that represent the complex numbers a, b, c, d and e , and where a is a positive real number. You are given that these five complex numbers are the roots of the equation $z^5 - a^5 = 0$.



(a) Justify each of the following statements.

- (i) A, B, C, D and E lie on a circle with centre O .

[1]

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© 2024 (ii) $ABCDE$ is a regular pentagon.

[2]

(iii) $b \times e^{\frac{2i\pi}{5}} = c$

[1]

(iv) $b^* = e$

[1]

(v) $a + b + c + d + e = 0$

[2]

- (b) The midpoints of sides AB , BC , CD , DE and EA represent the complex numbers p , q , r , s and t . Determine a polynomial equation, with real coefficients, that has roots p , q , r , s and t .

[3]

- 26 Draw the region in an Argand diagram for which $|z| \leq 2$ and $|z| > |z - 3i|$.

[3]

- 27 In this question you must show detailed reasoning.

The equation $f(x) = 0$, where $f(x) = x^4 + 2x^3 + 2x^2 + 26x + 169$, has a root $x = 2 + 3i$.
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- (a) Express $f(x)$ as a product of two quadratic factors.

[4]

- (b) Hence write down all the roots of the equation $f(x) = 0$.

[1]

28 Show that $\frac{5}{2-4i} = \frac{1}{2} + i$. [2]

29 Let $C = \sum_{r=0}^{20} \binom{20}{r} \cos r\theta$. Show that $C = 2^{20} \cos^{20}(\frac{1}{2}\theta) \cos 10\theta$. [8]



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30 The complex number $2 - i$ is denoted by z .

(i) Find $|z|$ and $\arg z$.

[2]

(ii) Given that $az + bz^* = 4 - 8i$, find the values of the real constants a and b .

[5]

31

(i) Sketch on a single Argand diagram the loci given by

(a) $|z| = 2$,

[2]

(b) $\arg(z - 3 - i) = \pi$.

[3]

(ii) Indicate, by shading, the region of the Argand diagram for which

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$$|z| \leq 2 \text{ and } 0 \leq \arg(z - 3 - i) \leq \pi.$$

[2]

32 In this question, give all non-real numbers in the form $re^{i\theta}$ where $r > 0$ and $0 < \theta < 2\pi$.

(i) Solve $z^5 = 1$. .[2]

(ii)

(iii) Hence, or otherwise, solve $z^5 + 32 = 0$. Sketch an Argand diagram showing the roots [4]



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33

- (i) Use de Moivre's theorem to show that

$$\sin 6\theta \equiv \cos\theta(6 \sin\theta - 32 \sin^3\theta + 32 \sin^5\theta).$$

[5]

- (ii) Hence show that, for $\sin 2\theta \neq 0$,

$$-1 \leq \frac{\sin 6\theta}{\sin 2\theta} < 3.$$

[7]

- 34 In an Argand diagram, the complex numbers $0, z$ and $ze^{\frac{1}{6}i\pi}$ are represented by the points O, A and B respectively.

- (i) Sketch a possible Argand diagram showing the triangle OAB . Show that the triangle is isosceles and state the size of angle AOB .

[4]

The complex numbers $1 + i$ and $5 + 2i$ are represented by the points C and D respectively. The complex number w is represented by the point E , such that $CD = CE$ and angle $DCE = \frac{1}{6}\pi$.

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- (ii) Calculate the possible values of w , giving your answers exactly in the form $a + bi$.

[5]

35

- (i) Use de Moivre's theorem to show that $\tan 4\theta \equiv \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$.

[4]

- (ii) Hence find the exact roots of $t^4 + 4\sqrt{3}t^3 - 6t^2 - 4\sqrt{3}t + 1 = 0$.

[5]

- (i) Solve the equation $z^6 = 1$, giving your answers in the form $re^{i\theta}$, and sketch an Argand diagram showing the positions of the roots.



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[4]

(ii) Show that $(1 + i)^6 = -8i$.

[3]

(iii) Hence, or otherwise, solve the equation $z^6 + 8i = 0$, giving your answers in the form $re^{i\theta}$.

[3]

37

(i) By expressing $\sin \theta$ in terms of $e^{i\theta}$ and $e^{-i\theta}$, show that

$$\sin^5 \theta \equiv \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta).$$

[4]

(ii) Hence solve the equation

$$\sin 5\theta + 4 \sin \theta = 5 \sin 3\theta$$

$$\text{for } -\frac{1}{2}\pi \leq \theta \leq \frac{1}{2}\pi.$$

[4]

38 The complex numbers 0, 3 and $3e^{\frac{1}{3}\pi i}$ are represented in an Argand diagram by the points O, A and B respectively.

(i) Sketch the triangle OAB and show that it is equilateral.

[3]

(ii) Hence express $3 - 3e^{\frac{1}{3}\pi i}$ in polar form.

[2]

(iii) Hence find $(3 - 3e^{\frac{1}{3}\pi i})^5$, giving your answer in the form $a + b\sqrt{3}i$ where a and b are rational numbers.

[3]



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(i) Use de Moivre's theorem to show that $\cos 5\theta \equiv 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$. [5]

(ii) Hence find the roots of $16x^4 - 20x^2 + 5 = 0$ in the form $\cos \alpha$ where $0 \leq \alpha \leq \pi$. [4]

(iii) Hence find the exact value of $\cos \frac{1}{10}\pi$. [3]

40

(i) Solve the equation $z^5 = 1$, giving your answers in polar form. [2]

(ii) Hence, by considering the equation $(z + 1)^5 = z^5$, show that the roots of

$$5z^4 + 10z^3 + 10z^2 + 5z + 1 = 0$$

can be expressed in the form $\frac{1}{e^{i\theta} - 1}$, stating the values of θ [5]

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41 Let $S = e^{i\theta} + e^{2i\theta} + e^{3i\theta} + \dots + e^{10i\theta}$.

(i)



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(a) Show that, for $\theta \neq 2n\pi$, where n is an integer,

$$S = \frac{e^{\frac{1}{2}i\theta}(e^{10i\theta} - 1)}{2i \sin\left(\frac{1}{2}\theta\right)}.$$

[4]



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(b) State the value of S for $\theta = 2n\pi$, where n is an integer.

[1]

(ii) Hence show that, for $\theta \neq 2n\pi$, where n is an integer,

$$\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos 10\theta = \frac{\sin\left(\frac{21}{2}\theta\right)}{2\sin\left(\frac{1}{2}\theta\right)} - \frac{1}{2}.$$

[3]

(iii) Hence show that $\theta = \frac{1}{11}\pi$ is a root of $\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos 10\theta = 0$ and find another root in the interval $0 < \theta < \frac{1}{4}\pi$.

[4]

42 The complex number z has modulus $2\sqrt{3}$ and argument $-\frac{1}{3}\pi$. Giving your answers in the form $x + iy$, where x and y are exact real numbers, and showing clearly how you obtain them, find

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(i) Z ,

[2]

(ii) $\frac{1}{(z^* - 5i)^2}.$

[5]

43 In an Argand diagram the points A and B represent the complex numbers $5 + 4i$ and $1 + 2i$ respectively.

(i) Given that A and B are the ends of a diameter of a circle C , find the equation of C in complex number form.

[4]

The perpendicular bisector of AB is denoted by l .

(ii) Sketch C and l on a single Argand diagram.

[2]



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(iii) Find the complex numbers represented by the points of intersection of C and l .



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[3]

44

(i) Use an algebraic method to find the square roots of the complex number $9 + 40i$. [6]

(ii) Show that $9 + 40i$ is a root of the quadratic equation $z^2 - 18z + 1681 = 0$. [1]



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- (iii) By using the substitution $z = \frac{1}{u^2}$, find the roots of the equation $1681u^4 - 18u^2 + 1 = 0$. Give your answers in the form $x + iy$, where x and y are real.

- 45 The complex number $x + iy$ is denoted by z . Express $3zz^* - |z|^2$ in terms of x and y .



[4]

[3]

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46 The loci C_1 and C_2 are given by $|z + 2| = 2$ and $\arg(z + 2) = \frac{5}{6}\pi$ respectively.

(i) Sketch, on a single Argand diagram, the loci C_1 and C_2 .

[4]



(ii) Find the complex number represented by the intersection of C_1 and C_2 .

[2]

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(iii) Indicate, by shading, the region of the Argand diagram for which

$$|z + 2| \leq 2 \text{ and } \frac{5}{6}\pi \leq \arg(z + 2) \leq \pi.$$

[2]

47

(i) Use an algebraic method to find the square roots of the complex number $5 + 12i$. You must show sufficient working to justify your answers. [5]

(ii) Hence solve the quadratic equation $x^2 - 4x - 1 - 12i = 0$. [5]

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- 48 The complex number $3 + ai$, where a is real, is denoted by z . Given that $\arg z = \frac{1}{6}\pi$, find the value of a and hence find $|z|$ and $z^* - 3$. [6]
- 49 Use an algebraic method to find the square roots of $11 + (12\sqrt{5})i$. Give your answers in the form $x + iy$, where x and y are exact real numbers. [6]

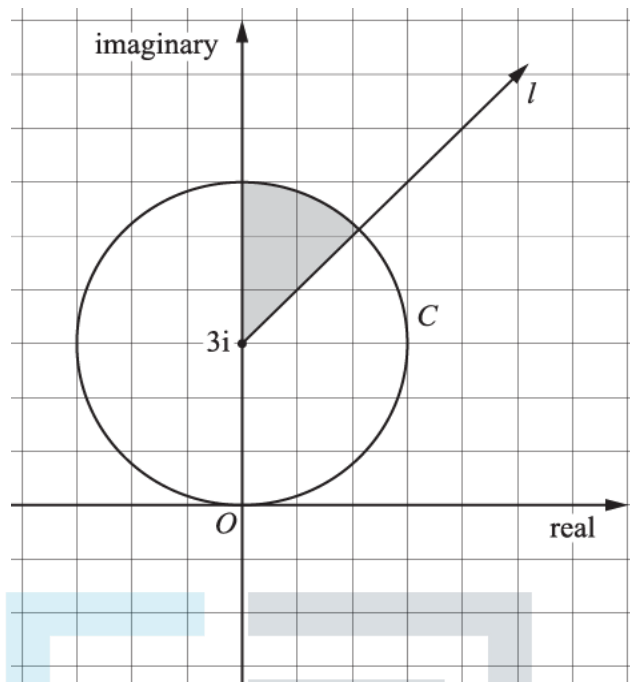


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The Argand diagram above shows a half-line l and a circle C . The circle has centre $3i$ and passes through the origin.

- (i) Write down, in complex number form, the equations of l and C

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[4]

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(ii) Write down inequalities that define the region shaded in the diagram. [The shaded region includes the boundaries.]

[3]

51 The complex number $7 + 3i$ is denoted by z . Find

(i) $|z|$ and $\arg z$,

[2]

(ii) $\frac{z}{4 - i}$, showing clearly how you obtain your answer.

[3]



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52 The loci C_1 and C_2 are given by $\arg(z - 2 - 2i) = \frac{1}{4}\pi$ and $|z| = |z - 10|$ respectively.

(i) Sketch on a single Argand diagram the loci C_1 and C_2 .

[4]

(ii) Indicate, by shading, the region of the Argand diagram for which

$$0 \leq \arg(z - 2 - 2i) \leq \frac{1}{4}\pi \text{ and } |z| \geq |z - 10|.$$

[3]



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